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Residential indoor air monitoring to assess long-term exposure to air pollution in susceptible populations for epidemiology: overview and early findings

Sun-Young Kim^{1*}, Kyumin Kim², Seokmin Ji¹, Miyoun Shin¹, Hyunji Kim¹, Young Shin Park¹, Soyoung Han³, Soyoun Bae³, Younghyun Kim⁴, Eehyun Park⁴, Jiyoung Hwang⁵, Seung-Hyun Cho⁶, Hee-Young Yoon⁷ and Jin Woo Song⁸

Abstract

While there has been accumulating evidence of the association between long-term exposure to air pollution and various health endpoints, most epidemiological studies relied on ambient air pollution which showed substantial deviation from indoor. This difference could lead to biased or imprecise health effect estimates, particularly for susceptible populations who spend most time indoors. Although a few studies focused on indoor air pollution, most monitoring was limited to one-time and/or relatively short-term periods. We aimed to provide an overview of an indoor monitoring campaign that was designed to assess long-term exposure to residential indoor PM_{2.5} for epidemiology and applied to 130 older adults with idiopathic pulmonary fibrosis in the Seoul Metropolitan Area, South Korea, for 2021–2023. Specifically, our design focused on (1) representation of long-term exposure, (2) identification of pollution sources, (3) assessment of infiltration, and (4) implementation of non-contact monitoring. Using RTI MicroPEMs, we measured concentrations of indoor PM_{2.5} and chemical components for five days including weekdays and weekends in four seasons as well as household information. We also collected matched outdoor samples to assess infiltration, and applied non-contact monitoring using postal mails for repeat visits. Our preliminary analyses of indoor hourly PM_{2.5} measurements (mean: 14.90 µg/m³ [standard deviation: 36.12]) showed temporal patterns with higher concentrations in winter (20.17 [50.54]) than in summer (11.19 [32.76]) and during the daytime than nighttime. Indoor concentrations were lower than outdoor (20.44 [SD: 61.63]) with consistent seasonal patterns but contrasted hourly patterns. Thirty-six chemical components of PM_{2.5} showed mostly similar temporal patterns to those of PM_{2.5} with lower concentrations and smaller differences between indoors and outdoors in summer than in other seasons. The infiltration factor characterized by median indoor to outdoor ratios of sulfur, nickel, and iron were 51–70%. Average PM_{2.5} concentrations tended to be high at households with cooking activities (15.14 vs. 10.96 µg/m³ at those without), humidifiers (19.19 vs. 13.16), pets (24.09 vs. 13.35),

*Correspondence:
Sun-Young Kim
sykim@ncc.re.kr

Full list of author information is available at the end of the article



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and smokers (16.76 vs. 11.79). Our indoor monitoring approach provides a practical guidance to future indoor monitoring studies that intend to assess the health effect of PM_{2.5}.

Keywords Cohort, Fine particulate matter, Health effect, Indoor air, Monitoring design

Introduction

While there has been accumulating evidence of the association between long-term exposure to air pollution including fine particulate matter (PM_{2.5}) and human health, one of the key limitations was indicated as the exposure assessment relying on ambient air pollution [38, 40]. Many epidemiological studies of long-term air pollution used ambient measurements mostly obtained from fixed monitoring sites without considering the level of exposure within domestic environments. Air pollution monitoring to assess indoor air quality across participants' homes has been sparse or limited to small sample sizes because of high compliance requirement of participants and relatively large burden of technicians [29, 30]. However, there was a good understanding of significant deviation of indoor air pollution from outdoor, since indoor air quality is affected by indoor microenvironments which are different from outdoor sources [3, 26, 33, 39]. Furthermore, as people spend 80–90% of their time indoors [19], the exposure assessment solely based on ambient measurements could result in substantial exposure measurement error. These differences may subsequently affect health effect analyses.

The impact of the difference in indoor exposure from outdoor on health analyses could be large for susceptible populations who stay longer indoors mostly at homes than the general population. Individuals with chronic lung diseases spent more time at homes than those without diseases in the same age groups and older individuals spent an ample majority of time indoors mostly at their homes [19, 22]. Large health effects of long-term ambient air pollution found in such populations could be even larger, when replaced with indoor air pollution in their domestic dwellings. However, most epidemiological studies of indoor air pollution employed short-term measurements and investigated temporary changes in lung function or disease status within a few days, weeks, or months [29]. Indoor air pollution assessed over extended periods can elucidate the role of air pollution in disease progression or health deterioration of susceptible populations.

The design of indoor monitoring is crucial to assess and characterize individual long-term exposure to indoor air pollution at people's residences, when the ultimate goal is the assessment of the health effect. Previous indoor monitoring of air pollution included several limitations in their designs to be overcome to achieve the goal. Indoor monitoring campaigns often collected samples for limited hours, days, and seasons, possibly because

of financial and/or logistical constraints, which may not represent the collective exposures over a long period. Although household information questionnaires are commonly applied to identify potential pollution sources such as cooking, cleaning, and housing conditions, a few characteristics successfully explained the pattern of indoor air pollution across households [33]. For PM_{2.5}, its chemical components related to specific sources can be a powerful resource to help the identification [1]. However, handy and affordable portable samplers frequently deployed to indoor monitoring mostly provide nephelometer measures of PM_{2.5} which do not allow the quantification of chemical components, as opposed to gravimetric measures. Another difficulty to assess indoor PM_{2.5} particularly in highly populated urban areas is the accurate quantification of infiltrated PM_{2.5}. Since indoor PM_{2.5} is primarily originated from outdoor PM_{2.5}, the amount of indoor PM_{2.5} infiltrated from outdoors needs to be distinguished from those emitted indoors [39]. In densely populated urban areas where people tend to live in domestic buildings, many studies located their samplers in gardens of buildings or indoor terraces of households where the outdoor measurements could be less comparable to corresponding indoor measurements [26, 49]. Lastly, indoor monitoring designs should adopt an approach to reduce visits to participants' homes which is one of the major barriers to participate in residential monitoring particularly when there are severe public health concerns such as COVID-19.

Leveraging a well-established cohort of participants aged mostly over 65 years and with idiopathic pulmonary fibrosis (IPF) in the Seoul Metropolitan Area, South Korea, we aimed to provide the overview of a monitoring campaign designed to assess the long-term exposure to indoor PM_{2.5} in a susceptible population for epidemiological application. Specifically, the design focused on the data collection to allow (1) representation of long-term exposure, (2) identification of potential pollution sources, (3) assessment of infiltration, and (4) implementation of non-contact sampling. These four focuses will help future studies accurately quantify individual exposures combined by indoor-originated and infiltrated PM_{2.5} over an extended period for subsequent health analyses. In this paper, we introduced the key features of our monitoring designs with respect to the four focuses and provided early results for PM_{2.5} measurements and household characteristics. Although this study was based on PM_{2.5} and participants with IPF, our indoor monitoring design can be applicable to future indoor monitoring studies

that intend health analyses of various air pollutants and populations.

Overview of indoor monitoring campaign

Study population

Our study population included 130 people who are a part of an idiopathic pulmonary fibrosis cohort, lived in the Seoul Metropolitan Area (SMA), visited one of the 12 tertiary hospitals in the SMA between 2021 and 2023 (Fig. 1; Table 1). The SMA is the fourth largest metropolitan area worldwide and consists of Seoul, Incheon, and Gyeonggi Province where 26 million population resides in 12,685 km² in 2020. Ambient PM_{2.5} concentrations in the SMA shows a strong seasonal variation with higher annual average concentrations (21 µg/m³ in 2020) compared to the 12 µg/m³ of the air quality standard proposed by the World Health Organization. IPF is a progressive lung disorder by scarring of the lungs from unknown causes, mostly prevalent in older males, with poor prognosis and an increasing pattern of incidence [21]. Given few identified risk factors other than smoking, recent studies of patients with IPF suggested air pollution as a potential risk factor of disease progression and mortality [4, 18, 31, 35, 44, 47] (Table S1). Air pollution could exacerbate disease progression through various biological mechanisms including oxidative stress,

telomere, and inflammation [36]. In the IPF cohort, participants visited hospitals every three or six months for monitoring their disease status and carried out medical examinations including pulmonary tests. These regular visits provide the unique opportunity to investigate the association of short-term as well as long-term exposure to air pollution with disease prognosis when we monitor the level of air pollution. The details of the study participants and their health data collection are provided elsewhere [48]. The study participants were 87% of males and the average age was 69.6 years with 93% of people greater than 60 years (Table 1). This study was approved by the Institutional Review Board of each site, including Asan Medical Center, Seoul, Republic of Korea (2021-0072).

IPF indoor monitoring campaign

We designed the IPF indoor monitoring campaign to measure PM_{2.5} concentrations within 130 homes of 130 participants for consecutive five days in four seasons from July, 2021 through August, 2023 with a subset also collecting outdoor PM_{2.5} measurements. The field team visited participants whose hospital visits were scheduled within the maximum four weeks and installed RTI MicroPEMs (RTI International, Research Triangle Park, NC, U.S.A.) in rooms where participants spend most of their time such as a living room or bedroom (Table S2, Figure S1). MicroPEMs operated continuously for five days at 0.4 L per minute and cycled 30 s on and 30 s off. The MicroPEM is a lightweight (~ 230 g) and quiet (3 dB above office background at 1 m) portable real-time PM sensor, which was widely applied in previous studies of indoor PM_{2.5} monitoring [9, 37]. Apart from most portable PM samples that provide optical measurements of PM_{2.5}, MicroPEMs concurrently collect filter samples using PTFE filters that provide gravimetric measurements, in addition to real-time optical measurements, and allow the quantification of PM_{2.5} chemical components. The gravimetric concentration was used to adjust the real-time PM mass concentration for improved accuracy [6]. We multiplied each real-time measurement by a correction factor, which is the ratio of the gravimetric-based PM_{2.5} concentration to the average value of all real-time concentrations. The adjusted real-time concentrations were averaged to a 1-minute scale for further analyses. The filter sample was subsequently analyzed for black carbon (BC), environmental tobacco smoke (ETS), and brown carbon (BrC) by 6-wavelength integrating sphere and 33 inorganic elements by X-ray fluorescence (XRF) at RTI international laboratories [20, 46]. The details of quantification and quality control procedure of PM_{2.5}, carbon species, and inorganic elements are provided in supplemental materials (See “S1. Details of quantification and quality control procedure of PM_{2.5} and of PM_{2.5} chemical components”). For the measurements

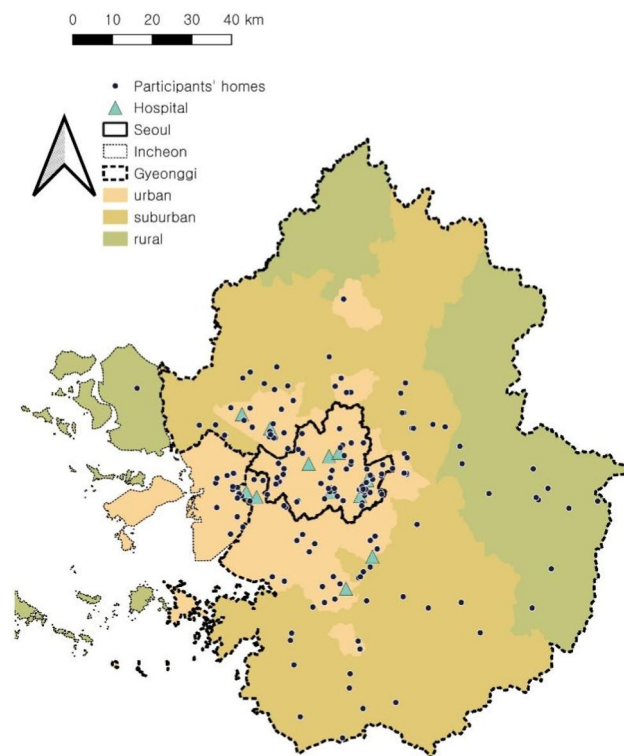


Fig. 1 Study area and 12 hospitals participating in the IPF indoor air pollution monitoring campaign in the Seoul Metropolitan Area, South Korea, from July 2021 through August 2023

Table 1 Summary of demographic and household characteristics of 130 participants from the IPF indoor air pollution monitoring campaign in the Seoul Metropolitan Area, South Korea, from July 2021 through August 2023

Variable	Value	130 participants with at least one out of four visits				95 participants with all four visits			
		N	%	Mean ^a	Median ^b	N	%	Mean ^a	Median ^b
Age (year)	40–59	9	6.9	13.40	10.23	3	3.2	11.41	7.29
	60–69	47	36.2	15.31	10.04	38	40.0	16.53	10.80
	70–79	69	53.1	15.25	11.53	51	53.7	16.23	12.46
	80–89	4	3.1	16.35	12.95	3	3.2	13.16	10.85
Sex	Male	107	82.3	15.48	11.13	80	84.2	17.01 ^c	12.20 ^c
	Female	22	16.9	13.69	10.01	15	15.8	11.26	8.29
Family members (person)	1	19	14.6	16.78	11.99	11	11.6	16.71	11.75
	2	62	47.7	13.75	10.00	48	50.5	14.24	10.43
	3	32	24.6	18.55	13.55	25	26.3	21.20	15.27
	4–6	16	12.3	12.06	8.13	11	11.6	12.02	8.05
Housing type	Single-family housing	17	13.1	15.65	11.91	10	10.5	16.55	13.04
	Multi-family housing	35	26.9	17.18	11.40	26	27.4	17.85	11.53
	High-rise apartment	76	58.5	14.15	10.56	59	62.1	15.25	11.36
House size	1–20	36	27.7	18.02	12.52	28	29.5	17.62	12.02
	21–40	71	54.6	13.60	9.98	49	51.6	14.80	10.89
	≥40	21	16.2	16.10	11.83	18	18.9	17.28	12.80
	No	2	1.5	11.09	7.18	1	1.0	8.37	5.10
Ventilation	Yes	127	97.7	15.24	11.00	94	99.0	16.18	11.65
	Window	108	83.1	15.53	11.29	80	84.2	16.73	12.09 ^c
Ventilation type	Window & machine	19	14.6	13.59	9.37	14	14.7	13.07	9.14
	<0.5	24	18.5	16.80	9.96	19	20	16.43	9.32
	0.5–1	20	15.4	12.29	8.71	11	12	13.88	9.63
	1–12	29	22.3	12.87	9.55	20	21	12.87	9.29
Air cleaner	≥12	38	29.2	16.04	12.47	30	32	17.85	14.04
	No	33	25.4	14.85	10.88	20	21.1	15.58	11.82
	Yes	96	73.8	15.29	10.96	75	78.9	16.24	11.52
	0	35	26.9	15.17	11.00	21	22.1	16.10	12.10
Number of air cleaners	1	64	49.2	16.41	11.80	51	53.7	17.68	12.53
	≥2	30	23.1	12.56	9.04	23	24.2	12.60	9.00
	<3	20	15.4	13.69	9.18	17	17.9	14.65	9.60
	3–9	23	17.7	16.79	10.99	15	15.8	18.87	11.49
Air cleaner Usage time (hour)	≥9	49	37.7	13.95	10.77	39	41.1	14.40	11.29
	No	85	65.4	13.16 ³⁾	9.66 ³⁾	64	67.4	13.61 ^c	10.07
	Yes	43	33.1	19.19	13.54	31	32.6	21.25	14.70

Table 1 (continued)

Variable	Value	130 participants with at least one out of four visits			95 participants with all four visits				
				PM _{2.5} mean			PM _{2.5} mean		
Cooking	No	7	5.4	10.96 ³⁾	8.27 ³⁾	6	6.3	11.31 ^c	8.63 ^c
	Yes	121	93.1	15.14	10.81	88	92.6	16.05	11.40
Cooking fuel	Natural gas (NG)	84	64.6	15.54	10.55	61	64.2	16.85	11.31
	Electricity	28	21.5	15.52	11.62	21	22.1	15.54	11.65
Cooking	NG & electricity	8	6.2	14.78	15.69	6	6.3	16.03	17.95
	0	7	5.4	10.96	8.27	6	6.3	11.31	8.63
Frequency (per day)	1–2	71	54.6	16.76	12.15	51	53.7	17.69	12.95
	3–4	50	38.5	12.84	8.91	37	38.9	13.79	9.27
Cooking time (min)	< 20	24	18.5	13.81	9.38	13	13.7	12.31	8.30
	20–40	53	40.8	15.35	12.02	42	44.2	15.92	12.47
Heating fuel	40–60	20	15.4	15.51	10.44	16	16.8	16.71	11.38
	≥ 60	24	18.5	15.69	9.89	17	17.9	18.59	11.16
Natural gas	None	5	3.8	16.85	10.92	2	2.1	22.34	12.61
	Natural gas	90	69.2	15.39	10.70	69	72.6	15.63	10.85
Oil	Oil	4	3.1	8.20	6.41	2	2.1	9.37	7.84
	Electricity	3	2.3	12.39	9.76	2	2.1	12.53	9.61
No	No	107	82.3	13.35 ^c	9.50 ^c	79	83.2	13.82 ^c	9.79 ^c
	Yes	22	16.9	24.09	17.97	16	16.8	27.37	20.43
Never smoker	Never smoker	28	21.5	11.79	8.88	21	22.1	11.02 ^c	8.26 ^c
	Former smoker	95	73.1	16.08	11.47	72	75.8	17.37	12.43
Current smoker	Current smoker	6	4.6	16.76	12.16	2	2.1	23.70	16.04
	No	121	93.1	15.07	10.86	89	93.7	16.03	11.54
Yes	Yes	8	6.2	16.84	12.11	6	6.3	17.19	12.24
	< 50 m (1 min walking)	40	30.8	14.14	10.89	30	31.6	15.38	12.04
50–100 m (2 min)	50–100 m (2 min)	33	25.4	16.09	10.70	24	25.3	16.37	10.46
	100–500 m (8 min)	46	35.4	16.19	11.48	36	37.9	17.09	12.16
≥ 500 m 8 min)	≥ 500 m 8 min)	9	6.9	11.33	9.59	5	5.3	11.99	10.09

^aMean of hourly PM_{2.5} measurements (ug/m³) over four seasons for each participant^bMedian of hourly PM_{2.5} measurements (ug/m³) over four seasons for each participant^c*p* < 0.1 in t-test or ANOVA

below the minimum detection limit (MDL), we replaced by the MDL divided by 2. MicroPEMs were also suitable for our indoor monitoring campaign given a relatively long operation without battery replacement up to five or six days to avoid additional requests to participants. The simple operation procedure of MicroPEMs allowed participants self-deploy devices for repeat sampling.

Key features of indoor monitoring design

Representation of long-term exposure

To assess long-term exposure and annual-average $PM_{2.5}$ concentration, we designed the IPF indoor monitoring campaign to collect continuous samples on different days and seasons. As shown in Table S3, previous studies that applied extensive indoor monitoring campaigns for epidemiological application were often limited to selective days and/or seasons. We carried out a five-day sampling for each of four seasons, including weekdays as well as weekends from Friday through Wednesday (Figure S2). The four seasons of spring, summer, fall, and winter were defined as March to May, June to August, September to November, and December to February (average temperature for 2021–2023: 12.8, 24.2, 14.9, and 0.3 °C, respectively). Before the visit to participants' homes on Friday for installation of MicroPEMs, our preparation began from Monday of the week. We made phone calls to each participant in order to notify our visit and confirm the visit time, prepared MicroPEMs including installation of filters and correction of flow rate, and packed all sampling items in the lab. One technician visited up to 5–6 participants within a day, given long driving distances between participant's homes with the maximum two driving hours (Table S2, Figure S4). We visited up to 11 participants a day depending on the recruitment of participants, the location of participants' homes, and the availability of technicians. Four repeated weekly-based sampling across four seasons for each of the 130 participants required more than two years of monitoring.

Implementation of non-contact sampling

Given that study participants are less inclined to accept repetitive visits, we received the sensors via postal mails. In particular, since our IPF monitoring campaign began in July 2021 in the middle of the COVID-19 pandemic, we developed a design that minimizes physical interactions. For successful implementation particularly for older adults, we provided the participants with detailed written instruction and follow-up phone calls (Figure S3). During the two days before the ending of the sampling on Wednesday, we made reservation for the shipping service and made phone calls to help participants pack devices for the return shipping (Figure S2). We also provided packaging materials such as mailing boxes, packing tapes, and bubble wraps. Starting from the third visit, we

switched completely to non-contact monitoring by mailing to participants' homes for installation in addition to participants' mails to return PM sensors.

Identification of pollution sources

$PM_{2.5}$ chemical components and household information questionnaire were used to identify potential sources of indoor air pollution. We constructed household information questionnaire composed of more than 100 questions based on previous studies of indoor monitoring [8, 11, 12, 28, 43]. The questions consisted of demographic features, housing conditions, behavioral characteristics, and indoor and outdoor environments (Table S3). For housing conditions, each participant provided the constructed year of a dwelling building, the size, floor level, and dividers of a residential dwelling, materials for walls and floors, and condensation or mold. Behavioral characteristics included heating, cleaning, smoking, and pet. Questions for Indoor and outdoor environments were related to ventilation, air cleaner, humidifier, and proximity to roads. We particularly intended to obtain the specific information on the characteristics that derived the PM changes in households of middle and high income countries: solid fuel burning, cooking, candle and incense, cleaning, and humidifier, in addition to smoking [42]. For example, the questions included the frequency, duration, and application type of cooking. Moreover, $PM_{2.5}$ chemical components, considered as surrogates of specific pollution sources, can help confirm major sources such as outdoor traffic, cooking, and cleaning. Although there were limited attempts that included $PM_{2.5}$ chemical components to indoor monitoring, a study in older adults with chronic obstructive pulmonary disease (COPD) showed indoor BC, considered as a marker of traffic, was associated with larger decline in lung function than indoor $PM_{2.5}$, suggesting the need of further investigation of indoor chemical components [14].

Assessment of infiltration

We selected 20% of participants living at the 5th floor or lower to obtain matched outdoor measurements in order to assess infiltrated $PM_{2.5}$ from outdoors. The matched outdoor and indoor measurements for the same participant can provide the indoor-outdoor (I/O) ratio, as the infiltration rate, for a specific $PM_{2.5}$ chemical component that is considered to be primarily of ambient origins with few indoor sources. For matched outdoor sampling, we placed MicroPEMs outside participants' homes with sampling devices hung on balcony bars or window bars, as opposed to gardens or within closed balconies often used in previous studies. This design allowed us to measure outdoor $PM_{2.5}$ to be comparable with indoor $PM_{2.5}$ at the same floor level as participants live. For the $PM_{2.5}$ chemical components to assess the infiltrated $PM_{2.5}$, we

considered sulfur, iron, and nickel which were adopted in previous studies of infiltration using I/O ratios as shown in Table S4 [2, 5, 17, 23, 24, 25, 34, 41, 45, 50]. We will identify the pollutants that satisfy the criteria to assess the infiltration [10, 24]: few known indoor sources, sufficient quantities in both indoor and outdoor environments, and I/O ratio equal to or less than one as well as smaller than $PM_{2.5}$ I/O ratio.

Data exploration

For the two years of monitoring period from July, 2021 through August, 2023 including nine 3-month periods that represent different seasons, we collected 579 $PM_{2.5}$ filter samples (62,130 hourly measurements) from 130 participants. These samples were composed of 448 indoor samples from all 130 participants, 105 outdoor samples from 35 (27%) participants, and 26 duplicated

samples from 25 (19%) participants (Table S5). In indoor monitoring for all 130 people, 95 participants (73%) completed all four visits. We excluded invalid samples resulting from instrument errors such as early stop and pump failure, and human errors such as filter damage, sample switch, lost real-time data file, and incorrect metadata (e.g. filter identification number).

$PM_{2.5}$

Table 2; Fig. 2 show the summary and temporal patterns of hourly-average $PM_{2.5}$ concentrations in 130 participants. The median and mean of indoor hourly $PM_{2.5}$ concentrations were 8.23 (interquartile range (IQR) = 10.44) and 14.90 (standard deviation (SD) = 36.12) $\mu\text{g}/\text{m}^3$, respectively, with the highest in the winter (median and mean: 9.19 and 20.17 $\mu\text{g}/\text{m}^3$, respectively) and lowest in the summer (7.04 and 11.19 $\mu\text{g}/\text{m}^3$) (Table 2, Figure S5).

Table 2 Summary statistics of hourly $PM_{2.5}$ concentrations ($\mu\text{g}/\text{m}^3$) by indoor and outdoor monitoring at homes of 130 participants from the IPF indoor air pollution monitoring campaign in the Seoul Metropolitan Area, South Korea, from July 2021 through August 2023

			N of obs.	N of samples	N of people	Min	Median	Max	IQR ^a	Mean	SD
Indoor	All		48,981	448	130	0.00	8.23	2877.25	10.44	14.90	36.12
	Day	Monday	9,902	430	130	0.00	8.51	1263.92	10.50	14.92	33.52
		Tuesday	8,840	395	128	0.00	7.75	826.70	10.22	14.20	32.27
		Wednesday	3,500	318	122	0.00	6.16	2877.25	8.27	13.50	63.11
		Thursday	551	44	39	0.01	4.22	57.19	6.45	8.34	9.55
		Friday	5,391	427	130	0.01	9.42	432.97	11.97	14.57	18.23
		Saturday	10,342	442	130	0.00	8.57	862.15	10.48	15.15	31.67
		Sunday	10,455	440	129	0.01	8.47	1199.25	10.31	16.22	40.76
	Season	Spring	12,147	109	109	0.00	8.91	805.09	9.68	15.47	32.25
		Summer	12,055	108	108	0.00	7.04	2877.25	10.15	11.19	32.76
		Fall	11,926	114	114	0.00	7.56	476.44	9.56	12.38	20.27
		Winter	12,853	117	117	0.00	9.19	1263.92	12.58	20.17	50.54
	Visit	1	13,916	127	127	0.00	8.47	1263.92	9.78	13.99	38.07
		2	12,314	113	113	0.00	8.02	829.34	10.21	14.12	24.77
		3	11,294	105	105	0.00	8.69	2877.25	10.32	17.76	47.09
		4	11,457	103	103	0.00	7.61	805.09	11.39	14.02	31.01
Outdoor	All		10,239	105	35	0.00	12.41	3079.93	15.16	20.44	61.63
	Day	Monday	1,935	88	34	0.02	12.50	1259.80	14.58	20.82	58.93
		Tuesday	1,377	73	33	0.05	12.37	1266.61	13.96	18.79	60.85
		Wednesday	530	44	27	0.00	8.20	3079.93	12.47	25.06	177.19
		Thursday	129	11	11	0.00	8.46	38.69	16.71	12.46	11.42
		Friday	1,393	104	34	0.04	12.47	420.29	15.55	18.97	27.56
		Saturday	2,490	104	34	0.03	13.43	1082.18	15.22	20.71	46.55
		Sunday	2,385	104	35	0.04	12.59	999.63	15.95	21.06	40.81
	Season	Spring	2,780	28	28	0.32	14.61	1266.61	16.32	28.35	84.29
		Summer	2,796	26	26	0.00	8.34	3079.93	9.89	12.72	77.62
		Fall	2,587	27	27	0.25	11.89	65.16	13.94	14.45	11.13
		Winter	2,076	24	24	1.63	18.16	157.48	24.31	27.71	26.35
	Visit	1	2,968	31	31	0.00	11.03	274.24	13.46	15.48	16.53
		2	3,068	29	29	0.11	12.66	1266.61	18.40	26.11	81.00
		3	2,002	22	22	0.51	14.11	3079.93	14.62	23.36	92.67
		4	2,201	23	23	0.02	12.46	321.99	14.04	16.55	15.73

^aInterquartile range

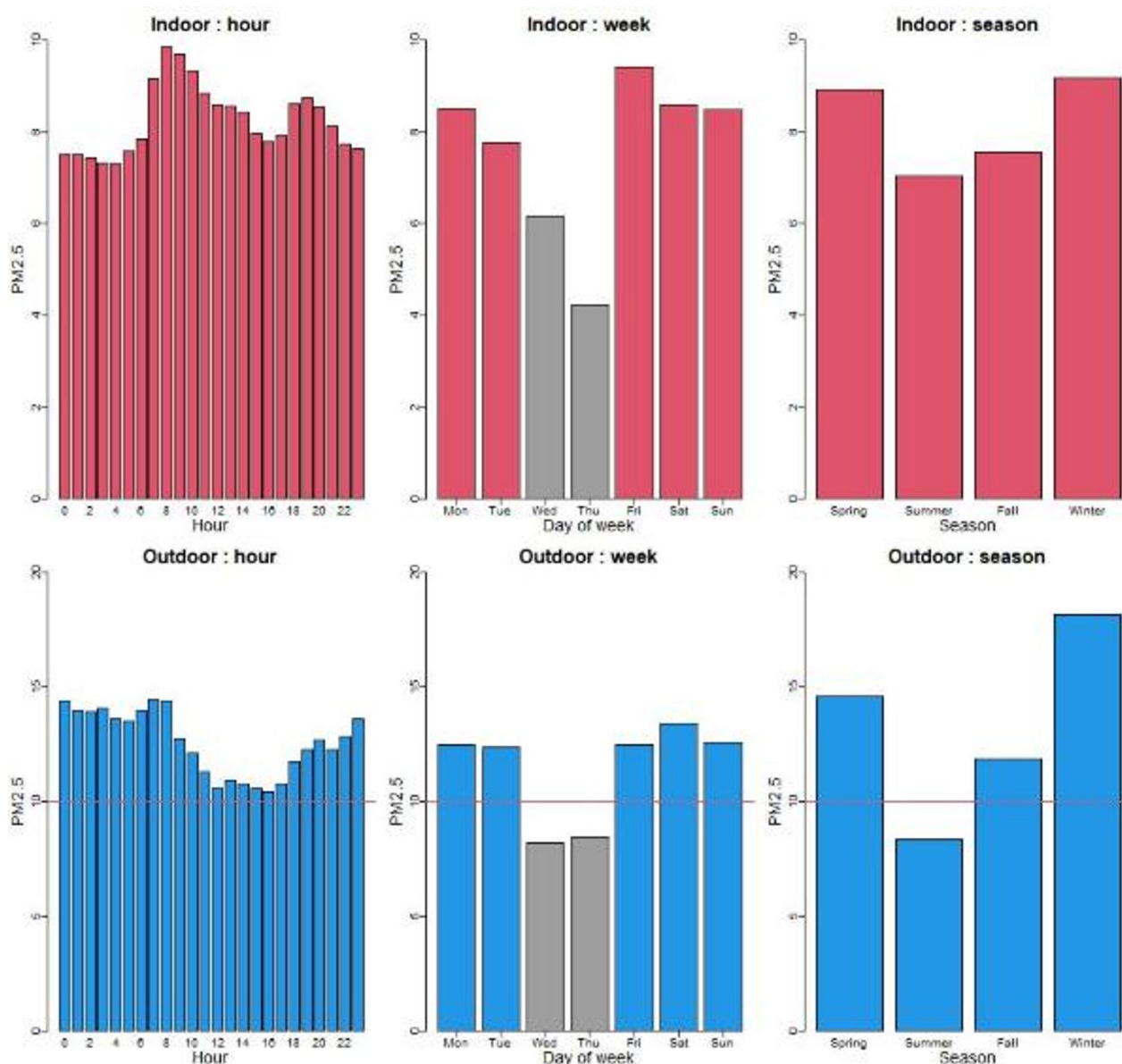


Fig. 2 Patterns of medians of PM_{2.5} hourly concentrations across hours, days of the week, and seasons by indoor (130 people) and outdoor (35 people) monitoring samples from the IPF indoor air pollution monitoring campaign in the Seoul Metropolitan Area, South Korea, from July 2021 through August 2023 (gray colors for Wednesday and Thursday including the return shipping of sampling devices)

This seasonal contrast indicates the impact of the ambient air quality affected by constant heating in cold winter and frequent rain in humid summer. To investigate the representative long-term average concentrations depending on the season, we compared 1- to 3-season averages to 4-season averages in 95 people who provided all four seasonal samples. The investigation showed relatively small root mean square error for most 2-season averages (6.41–6.87 µg/m³ for spring-summer, spring-fall, summer-winter, and fall-winter) comparable to those for 3-season averages (4.43–6.99 µg/m³). This comparability suggests the inclusion of at least two seasons with

one high- and one low-concentration days to represent long-term average concentrations (Figure S7). Daily average concentrations were similar between weekdays and weekends, suggesting stable time-activity patterns of older adults who were mostly retired. Hourly patterns showed higher concentrations during the daytime than the nighttime. In particular, morning and evening hours (7–10 a.m. and 6–8 p.m.) including cooking and cleaning activities gave high concentrations. Across four visits, indoor PM_{2.5} concentrations were more variable between households than within a household (Figure S6). Outdoor PM_{2.5} concentrations were higher and more variable

(median [IQR] and mean [SD] = 12.41 [15.16] and 20.44 [61.63] $\mu\text{g}/\text{m}^3$) compared to indoor $\text{PM}_{2.5}$ (Table 2). Although daily and seasonal patterns were similar between outdoor and indoor $\text{PM}_{2.5}$, the hourly pattern was reverse with higher outdoor concentrations in the nighttime (Fig. 2). These patterns were consistent when we restricted our analysis to 35 people who provided both indoor and outdoor measurements (Figure S8).

$\text{PM}_{2.5}$ chemical components

Carbonaceous components of 460 samples showed relatively higher mean concentrations (0.57 [SD: 0.33], 1.27 [0.84], and 2.01 [4.88] $\mu\text{g}/\text{m}^3$ for BC, BrC, and ETS, respectively) than 33 inorganic elements where sulfur gave the highest concentration (0.89 [0.72] $\mu\text{g}/\text{m}^3$) (Table S6, Figure S7). Across four seasons, all $\text{PM}_{2.5}$ components generally showed similar patterns with lower concentrations in the summer than the other seasons (Figure S9). Carbon and inorganic species comprised 50% of $\text{PM}_{2.5}$ mass on average with some seasonal differences (48, 34, 60, and 57% on average for the spring, summer, fall, and winter, respectively). When we compared matching indoor and outdoor measurements of 35 participants, most $\text{PM}_{2.5}$ chemical components showed higher concentrations in outdoors than indoors with larger differences in the winter than the other seasons (Table S6). The median I/O ratios of sulfur, nickel, and iron which are commonly used for assessing infiltration factor were ranged between 0.51 and 0.70, indicating 51–70% of infiltrated $\text{PM}_{2.5}$ from outdoor air depending on the elemental tracer (Table S7).

Household information characteristics

About 85% of 130 participants spent more than 15 h at home and almost all participants performed regular ventilation by opening windows (98%) (Table 2). Three fourth of the participants used air cleaners (74%), while one third had humidifiers (33%). Most participants cook their meals at homes (93%) with cooking fuel of natural gas (92%). The most common heating fuel was also natural gas (69%). One fifth of participants had pets (17%). About 5% of participants smoked or had smokers in family members. Average $\text{PM}_{2.5}$ concentrations were higher at 130 households with cooking activities (15.14 vs. 10.96 $\mu\text{g}/\text{m}^3$), humidifiers (19.19 vs. 13.16), pets (24.09 vs. 13.35), and smokers (16.76 vs. 11.79) than those without (Table 1). Although $\text{PM}_{2.5}$ concentrations were similar between the households with and without air cleaners (15.29 and 14.85, respectively), those having multiple air cleaners tended to show lower concentrations (12.56) compared to those having no or one cleaner (15.17 and 16.41, respectively). Households having ventilation behaviors using both open windows and machines showed lower $\text{PM}_{2.5}$ (13.39) compared to those using

open window only (15.53). These patterns were generally similar to those in 95 households that provided complete measurements across four seasons.

Discussion

This paper illustrated our intention to design a monitoring campaign with the specific interest in assessing the long-term residential exposure to indoor $\text{PM}_{2.5}$ and identifying factors that influence indoor air quality for investigating the health effect. This targeted interest shaped our indoor monitoring campaign to extend temporal coverage to represent long-term exposure and to identify sources of indoor $\text{PM}_{2.5}$ composed of indoor and ambient origins. Specifically, our monitoring design included different hours, days, and seasons that will allow us to compute annual-average indoor $\text{PM}_{2.5}$ concentrations representative to each individual. Including $\text{PM}_{2.5}$ chemical components, home questionnaires, and matching outdoor-to-indoor $\text{PM}_{2.5}$ monitoring will help distinguish between indoor-generated and infiltrated $\text{PM}_{2.5}$, and characterize total indoor $\text{PM}_{2.5}$.

Our study is aligned with the emerging research interest in the health effect of long-term exposure to indoor air pollution. Given the extended time spent indoors and the reported difference in indoor air pollution from outdoor, there is a growing interest in studying the health effect of indoor air pollution. However, most of these studies examined relatively short-term changes over a day or a week in indoor air pollution. A study in Massachusetts, U.S. reported that increased weekly average concentrations of indoor $\text{PM}_{2.5}$ and BC were associated with decreased lung function on the following week in 125 COPD patients with stronger associations than those of outdoor concentrations [14]. Another U.S. study for 44 children also showed the association of indoor $\text{PM}_{2.5}$ for the previous 48 h with lung function decline but no association of outdoor [16]. Building upon the evidence of the association of long-term exposure to air pollution with disease development and progression relying on ambient air pollution, recent studies attempted to look at the same association by replacing with indoor air pollution. However, their monitoring was often based on limited temporal coverage such as a single day across two seasons or each month [11, 28]. Recently, the Subpopulations and Intermediate Outcome Measures in COPD and AIR (SPIROMICS AIR) study applied an extended indoor monitoring period over two weeks but limited to two or three seasons, instead of all four, for each of 216 adults from 12 clinical centers in the continental U.S. Hansel et al., [12]. Furthermore, based on the indoor monitoring data along with their household information questionnaire and infiltration rate obtained from another study, the SPIROMICS AIR developed a prediction model (cross-validation R^2 of 0.45) to estimate indoor $\text{PM}_{2.5}$ concentrations for

more than two thousands participants who did not have indoor PM_{2.5} measurements and investigated the association with lung function [13, 51]. More studies need to help elucidate the role of exposure to indoor air pollution for extended periods in disease progression and diverse populations and environments.

Our well-designed monitoring can contribute to our future work where we can address longstanding questions related to indoor air pollution assessment. First, we plan to investigate cost-effective monitoring designs that contain reduced hours, days, or seasons and provide the long-term exposure estimates comparable to those of the complete design including the extended temporal coverage as our study. Our suggestion of cost-effective monitoring designs can minimize efforts and resources with the maximized data quality. Second, we will identify the most responsible indoor sources. In the present study, we found little differences in indoor PM_{2.5} concentrations between households according to their air cleaner usage and cooking activities as opposed to previous findings in high-income countries [42]. Since these little differences could be due to the impact of other individual and/or household characteristics, we will apply regression approaches to investigate the association with potential indoor sources after accounting for such characteristics. PM_{2.5} chemical components will also help improve our understanding of potential sources, when they are applied to source apportionment approaches [7, 15]. Third, we will quantify infiltrated PM_{2.5} relying on the ratios of a specific PM_{2.5} component which best satisfies the criteria after the investigation of candidate components used in previous studies. Fourth, given infiltration and indoor sources as well as our previously-validated prediction models which allow us to estimate outdoor concentrations at people's homes, we will develop a prediction model to estimate individual indoor concentrations of PM_{2.5} for people without indoor measurements.

In addition to the four focuses intended in our monitoring designs, this study has some additional design suggestions to future studies of indoor air pollution monitoring. Although our study was dedicated to PM_{2.5} and patients with IPF, our monitoring design targeting long-term exposure assessment for health analyses can be easily extended to other susceptible populations such as asthmatic children and cancer patients as well as the healthy population. In addition, we did not apply the emerging monitoring approach using low cost sensors with online data transfer service [27, 32]. Although these devices are powerful tools to obtain data without individual visits leading to the substantial reduction in logistical efforts, internet service was frequently not accessible or available in elderly households. Future studies should consider unique features of monitoring environments

related to population characteristics in their choices for indoor monitoring.

In summary, recent increasing interests in the health effect of individual long-term exposure to indoor air pollution require active data collection in order to overcome the lack of monitoring data. However, the logistical and financial difficulties in indoor monitoring seek effective solutions in designs that help achieve data accuracy, understand the nature of exposure, and minimize the burdens of participants and technicians. Our practical guidance can provide such solutions and allow further epidemiological investigation.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12940-026-01334-3>.

Supplementary Material 1.

Human ethics and consent to participate

Consent obtained directly from participants.

Authors' contributions

Sun-Young Kim: Conceptualization, Formal analysis, Funding acquisition, Methodology, Visualization, and Writing — original draft. Kyumin Kim: Conceptualization, Data curation, Methodology, and Resources. Seokmin Ji: Data curation, Investigation, and Methodology. Miyou Shin, Methodology, Data curation, Investigation, and Methodology. Hyunji Kim: Project administration, Resources. Young Shin Park: Data curation, Investigation, Methodology, Validation, and Writing — review & editing. Soyoung Han: Data curation, Investigation, Methodology, and Validation. Soyoung Bae: Data curation, Investigation, and Validation. Younghyun Kim: Data curation, Investigation, and Validation. Eehyun Park: Data curation, Investigation, and Validation. Jiyoung Hwang: Data curation and Investigation. Seung-Hyun Cho: Conceptualization, Methodology, and Writing — review & editing. Hee-Young Yoon: Conceptualization, Resources, and Writing — review & editing. Jin Woo Song: Conceptualization, Methodology, and Writing — review & editing. All the authors have read and agreed with the published version of the manuscript.

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Data availability

Our research data is unavailable to access because the data includes confidential information of cohort participants.

Declarations

Ethics approval and consent to participate

This study involves human participants and the study was approved by the Institutional Review Board of Asan Medical Center (2021-0072). Participants gave informed consent to participate in the study before taking part.

Competing interests

The authors declare no competing interests.

Author details

¹Department of Public Health and AI, National Cancer Center Graduate School of Cancer Science and Policy, Goyang-si, Gyeonggi-do 10408, Republic of Korea

²Department of Health Administration, Gyeonggi University of Science and Technology, Siheung-si, Republic of Korea

³Department of Statistics and Data Science, Yonsei University, Seoul, Republic of Korea

⁴Department of Statistics, Sungkyunkwan University, Seoul, Republic of Korea

⁵Gangarosa Department of Environmental Health, Emory University, Atlanta, GA, USA

⁶Center for Exposure and Protection, Research Triangle Institute, Research Triangle Park, NC, USA

⁷Department of Pulmonary and Allergy, Soonchunhyang University Hospital, Seoul, Republic of Korea

⁸Department of Pulmonary and Critical Care Medicine, Asan Medical Center, Seoul, Republic of Korea

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